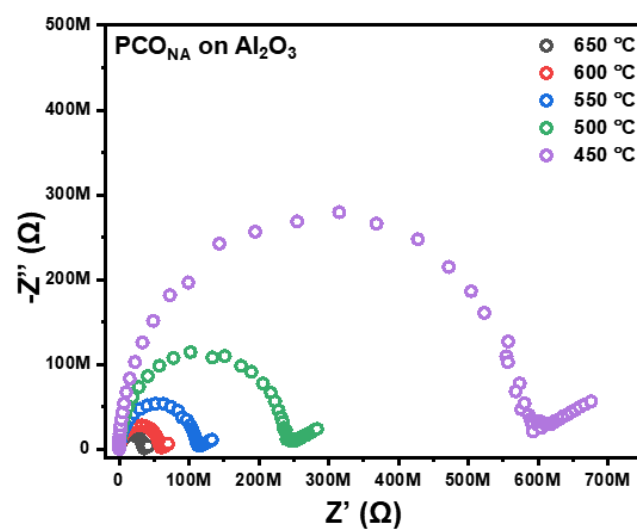
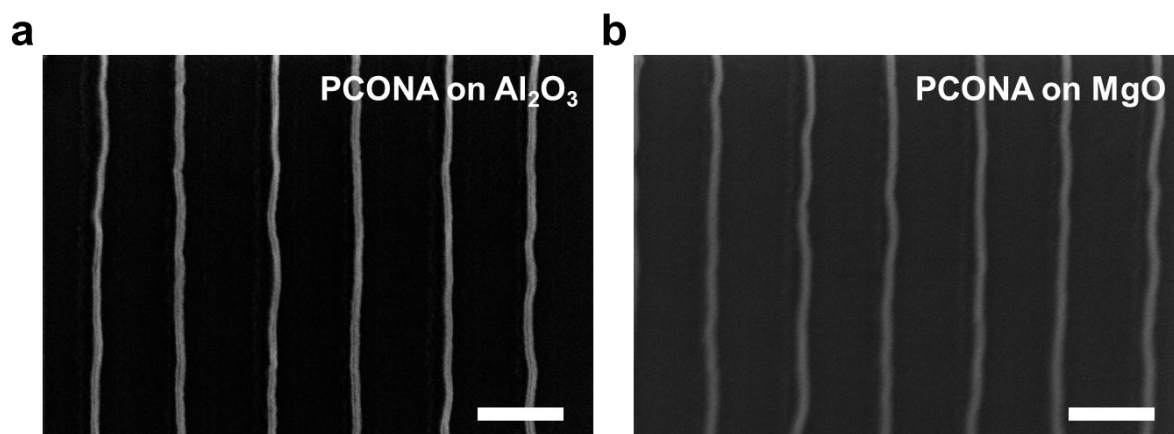
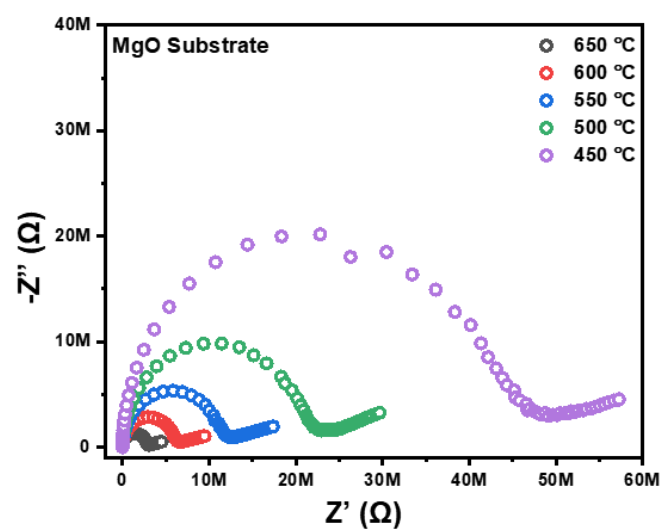




Supplementary Fig. 3 Impedance spectra measured from 450 °C to 650 °C on uninfiltrated PCO_{NA} fabricated on Al₂O₃ insulating substrate.



Supplementary Fig. 4 Scanning electron microscopy (SEM) images of PCO_{NA} fabricated on **a** Al_2O_3 and **b** MgO substrates, respectively (scale bar, 200 nm).



Supplementary Fig. 5 Impedance spectra measured from 450 °C to 650 °C on uninfiltrated PCO_{NA} fabricated on MgO insulating substrate.